



AgeisFIM

Project Report

Submitted By:

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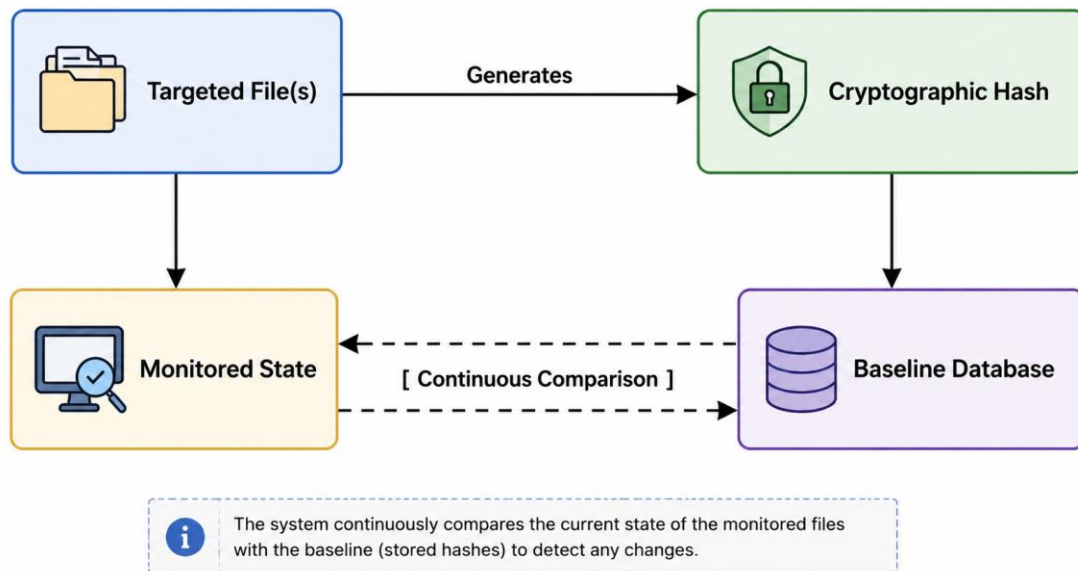
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1. Introduction

AgeisFIM is an advanced File Integrity Monitoring system designed to detect unauthorized changes in critical files and directories. The system continuously scans files, generates cryptographic hashes, and compares them with stored baseline values. Any unexpected change triggers alerts and logging mechanisms to help administrators maintain system integrity and security.



2. Problem Statement

Modern systems are vulnerable to malware, insider threats, and unauthorized file modifications. Traditional antivirus systems may fail to detect silent changes in sensitive files. Organizations therefore require a dedicated monitoring system capable of detecting additions, deletions, and modifications in real time.

3. Objectives

The main objectives of AgeisFIM are:

- Detect unauthorized modifications in files.
- Monitor file creation, deletion, and renaming.
- Maintain integrity using SHA-256 hashing.
- Generate alerts and activity logs.
- Improve overall cybersecurity posture.

4. Scope of the Project

The project focuses on monitoring selected files and directories within a computer system. It compares file hashes periodically and stores activity logs for analysis. The system does not recover deleted files or automatically block attacks, but it provides early warning mechanisms for suspicious activity.

5. System Architecture

The AgeisFIM architecture consists of four main components:

1. File Scanner – Reads monitored files.
2. Hash Generator – Generates SHA-256 hashes.
3. JSON Baseline Manager – Stores baseline hashes locally in JSON format.
4. Alert Module – Sends notifications when changes are detected.

These components work together to ensure secure monitoring and accurate detection.

6. Working Mechanism

The user selects files or directories for monitoring. The system calculates baseline hashes and stores them locally in JSON format inside the project folder. During monitoring, the application continuously scans the files and compares newly generated hashes with baseline hashes. If differences are found, the system records the activity and generates alerts.

The core of the detection engine relies on one-way cryptographic hashing functions. The Secure Hash Algorithm 2004 variant (SHA-256) is chosen as the primary cryptographic standard due to its high collision resistance. When a file stream is processed, it is read in binary blocks to handle large volumes without consuming excessive RAM.

The mathematical mapping can be expressed as:

$$\mathbf{H} = \text{SHA-256}(\mathbf{M})$$

Where **M** represents the sequence of binary blocks within the target file, and **H** is the resulting fixed-size 256-bit (64-character hexadecimal) digest value. Because cryptographic hashes change significantly even with minor adjustments to the underlying data (the avalanche effect), a single altered bit in a massive database configuration file instantly changes the resulting hexadecimal string, triggering an alert.

7. Methodology

The methodology used in AgeisFIM includes:

- Collecting file metadata.
- Generating cryptographic hashes.
- Storing baseline values.
- Performing continuous monitoring.
- Comparing current and baseline hashes.
- Logging and reporting suspicious activities.

This methodology ensures reliability and minimizes false positives.

8. Security Features

AgeisFIM provides multiple security features including secure hash verification, log management, real-time monitoring, and suspicious activity alerts. The use of SHA-256 prevents attackers from easily manipulating monitored data. Logs can also assist in forensic investigations after security incidents.

9. Tools and Technologies

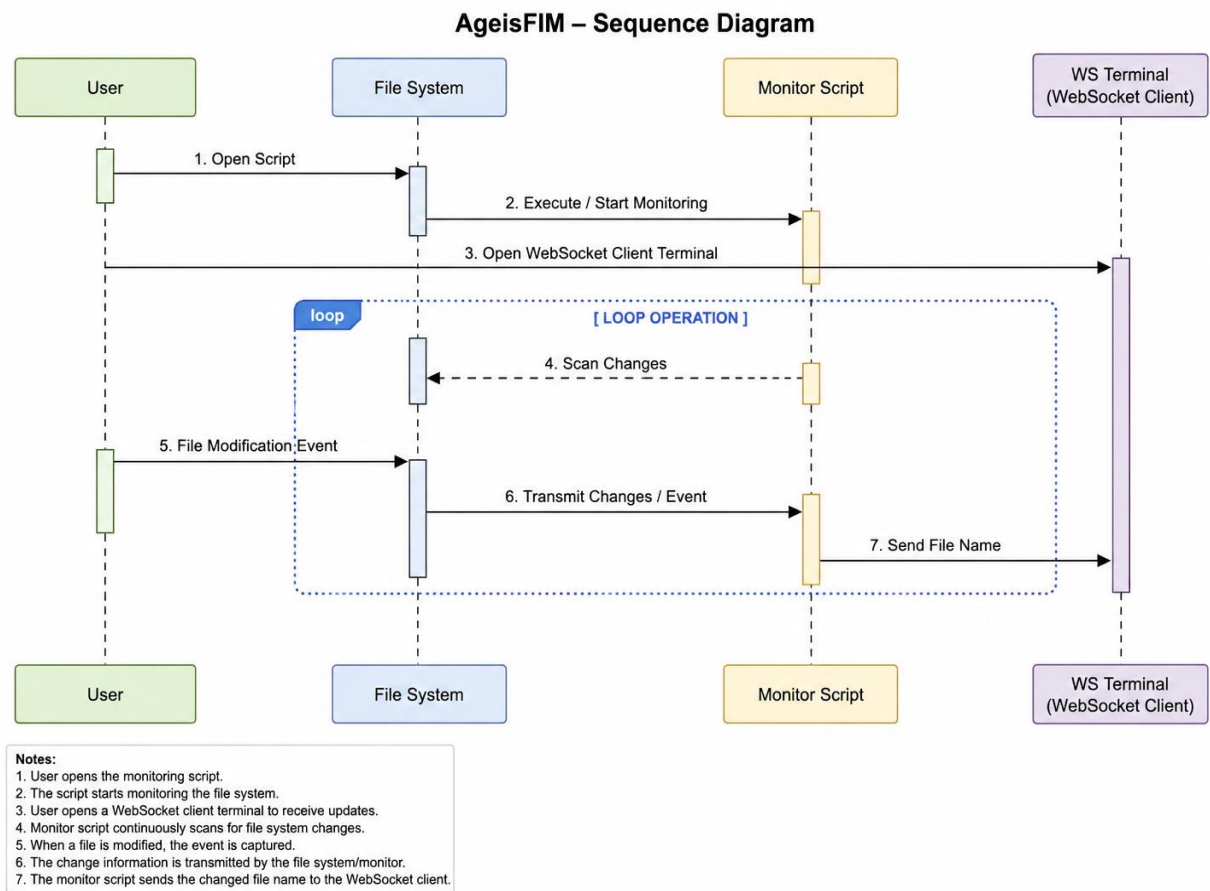
The following tools and technologies are used in the development of AgeisFIM:

- Python – Core programming language.
- SHA-256 – Hashing algorithm.
- Visual Studio Code – Development environment.
- Flask – Backend API framework.
- Flask-CORS – Enables communication between frontend and backend.
- JSON – Stores baseline hashes locally.
- HTML/CSS/JavaScript – Frontend UI development.

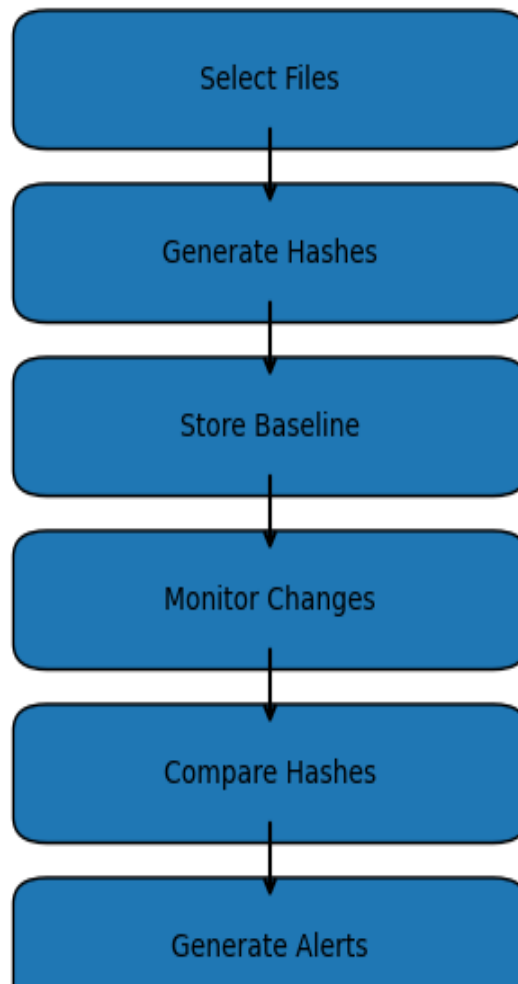
10. UML and System Diagrams

The following diagrams explain the internal workflow and data communication within AgeisFIM.

Sequence Diagram



System Flow Diagram



11. Software Requirements

- Python 3.10 or higher
- Required Libraries: Flask, Flask-CORS

12. Advantages and Limitations

Advantages:

- Improves system security.
- Detects unauthorized modifications quickly.
- Maintains audit trails.
- Lightweight and efficient.

Limitations:

- Cannot recover deleted files.
- Does not automatically block attacks.
- Requires proper baseline management.

14. Conclusion

AgeisFIM is an effective cybersecurity solution for detecting unauthorized file modifications and maintaining data integrity. By combining cryptographic hashing with continuous monitoring, the system helps organizations detect suspicious activity and strengthen overall system security.

15. References

- GitHub Repository: <https://github.com/abdulwasay-lab/File-Integrity-Monitor.git>
- Online Available Tools: <https://www.dnsstuff.com/file-integrity-monitoring-fim-in-cybersecurity>
- Wikipedia: https://en.wikipedia.org/wiki/File_integrity_monitoring